

AVISH COORAY

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EDUCATION

Iowa State University - Ames, IA

B.S. Data Science & Computer Science | Minor in Artificial Intelligence | Honors Program

Expected Graduation: Spring 2029

Aug 2025 - Present

Cumulative GPA: 4.0

SLIIT International - Colombo, Sri Lanka

Certificate in Computer Science & AI Foundations (120 hours)

Nov 2024

Pearson Edexcel International Advanced Level

GPA equivalent: 4.0 • June 2025

SKILLS

Programming: Python, SQL, Java, HTML, CSS, JavaScript

Libraries & Tools: pandas, numpy, scikit-learn, matplotlib, seaborn, PyQt5, OpenMDAO, pytest, Git, VS Code, IntelliJ, Jupyter

Concepts: Data pipelines, API integration, machine learning, Gaussian processes, HPC deployment, data visualization

RESEARCH EXPERIENCE

Data Science Researcher | NexGen PPT Lab - Iowa State University

Jan 2026 - May 2026

- Architected a digital twin for firefighter PPE smoke characterization, integrating FDS fire simulation, ANSYS Fluent particle tracking, and a Python ML pipeline across 140+ physical burn experiments; improved GPR surrogate R^2 from 0.026 to 0.759 (RMSE 22.1M #/cm³) via composite ARD kernel, one-hot fuel encoding, and experiment-grouped train/test split
- Built a modular 6-module data pipeline fusing OPS and SMPS particle sensor outputs with cone calorimeter time series, handling instrument-specific quirks (10× dilution corrections, transposed layouts, 60s bin alignment) and metadata filtering to produce a clean 4,955-row, 23-feature training set from 105 validated experiments
- Deployed full ML pipeline to Iowa State's Nova HPC cluster via VS Code tunnel; conducted multi-model suitability analysis, identifying GPyTorch SVGP (GPU-accelerated) as the priority upgrade as the dataset scales past sklearn's memory limits

Data Science Researcher | Translational AI Center (REU) - Iowa State University

June 2026 – Present

- Developing ML-based deconvolution methods to reconstruct true nanoscale surface geometry from degraded Atomic Force Microscopy (AFM) images, accounting for tip wear as a convolution artifact
- Investigating inverse problem approaches to estimate AFM tip degradation state from image outputs, enabling real-time correction of measurement error in nanoscale surface characterization

TECHNICAL EXPERIENCE

Data & Software Contributor | PrISUm Solar Car Team - Iowa State University

Sept 2025 - Present

- Built a PyQt5 simulation interface for race strategy. Rewrote the inherited UI, developed a wrapper class integrating an OpenMDAO speed optimizer with solar irradiance inputs, and visualized energy-over-distance output for route planning
- Automated geospatial data processing by building a KML → CSV parsing pipeline and refactoring the coordinate-matching workflow to resolve floating-point precision issues, improving route/weather data alignment reliability
- Refactored Baron Weather API threading architecture to support progress tracking; improved data validation performance by replacing nested loop comparisons with dictionary-based lookups ($O(n^2) \rightarrow O(n)$)

LEADERSHIP

House President | Converse House, Friley Hall - Iowa State University

Sept 2025 - May 2026

- Lead a residential community of 30 students through event planning and engagement initiatives
- Represent resident concerns in the hall senate and collaborate with RAs and hall leadership on community decisions

Dining Supervisor | Union Drive Marketplace - Iowa State University

Sept 2025 - Present

- Promoted to supervise 5-10 student employees in a high-volume dining operation, enforcing standardized workflow procedures and training new hires
- Manage complex employee situations and communicate workflow improvements directly to management

Volunteer IT Teacher | Boys Industrial Home & Orphanage - Colombo, Sri Lanka

July 2022 - Sept 2023

- Delivered weekly computer literacy lessons covering file management, Microsoft Office, email communication, and internet safety to students across varied learning levels
- Introduced basic graphic design concepts as an extension of the core curriculum